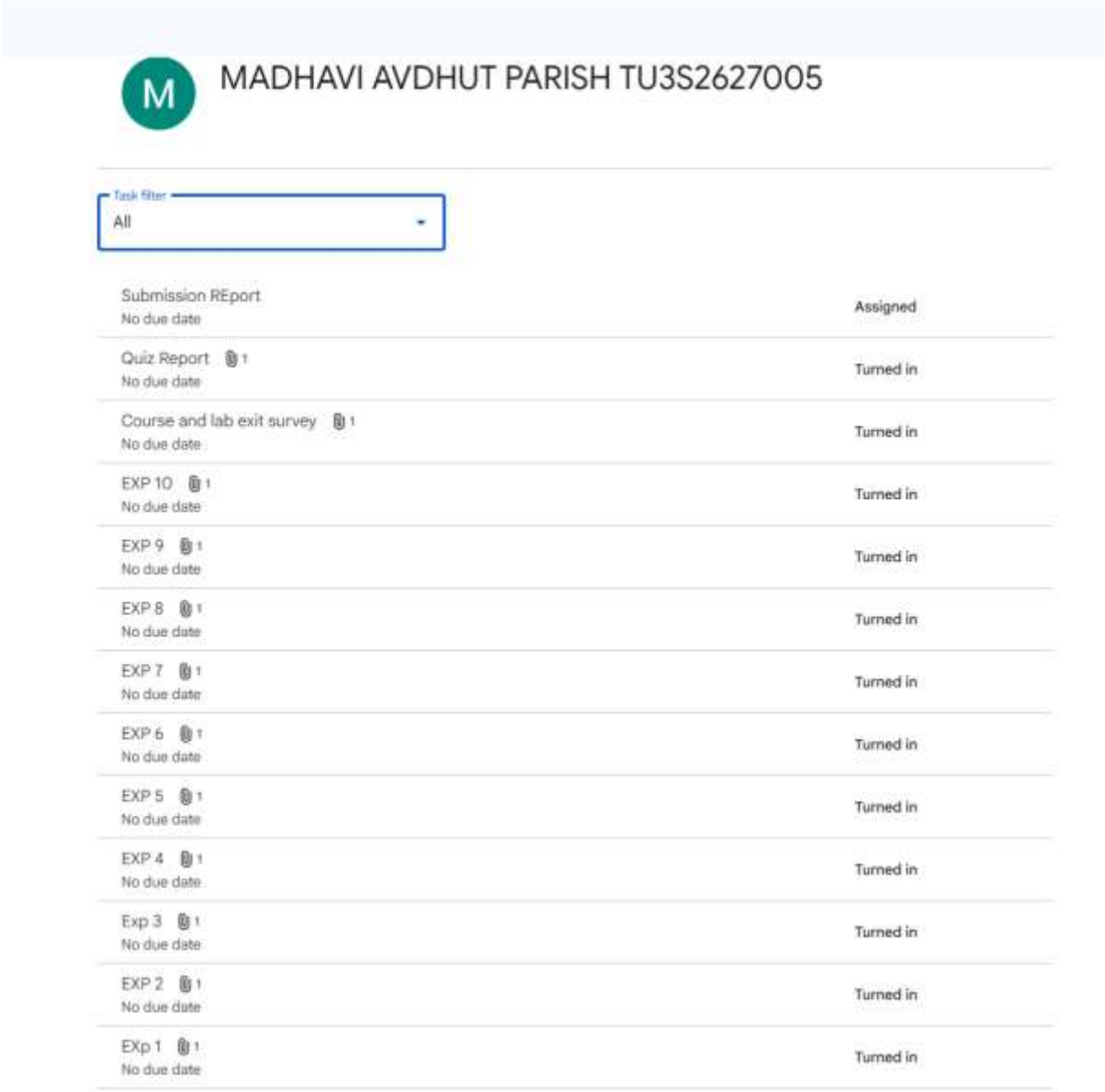
Submission Report



Avdhut Pavish madhavi C2 Batch
C-60 SE TU3526270005

Terna Engineering College, Nerul

AIPS

Assignment No. 1

8/10

1. The recurrence relation of an algorithm is as given below.

$T(n) = T(n/2) + 1$ for $n > 1$

$= 1$ for $n = 1$

use Backward substitution method to find out the equation for the time complexity of the algorithm write the final answer in Big O notation.

→ By Backward substitution method

put $n = n/1$

$T(n) = T(n/2) + 1$ — (1)

$T(n) = T(n/4) + 1$ — [2]

Substituting eqn 2 in 1

$T(n) = T(n/4) + 1 + 1$ — [3]

similarly

$T(n/4) = T(n/8) + 1$ — [4]

substituting eq 4 in 3

$T(n) = T(n/8) + 3$

similarly

$T(n) = T(n/16) + 4$

After K substitutions

$T(n) = T(n/2^K) + K$

$\frac{n}{2^K} = 1$

$n = 2^K$

$K = \log_2 n$

Avdhut Parish Madhavi
SE-C2 C60 TU3526270005

Terna Engineering College, Nerul

8/10

Assignment No. 2

Q1. → N-Queen Problem and Sum of Subsets

A) N-Queen problem using backtracking

The N-Queen problem is to place 8 queen on an 8x8 chessboard so that no two queen attack each other. No two queen can be in the same row, colown or diagonal.

Backtracking approach:

1. Place one queen in the first row
2. Check whether the chosen position is safe
3. If safe, move to the next row and place another column.
4. In no safe position is available, remove the previous queen and try another
5. Repeat until all 8 queen are placed.

The column positions for rows 1 through 8 are:

[1, 5, 8, 6, 3, 7, 2, 4]

Here, the number in each position represents the column containing the queen in that row

Course Exit Survey - 2023102

2023102 - Algorithms for Intelligent Problem Solving | Division C, Sem 3

Admission Number *

TU3S2627005

Verified

MADHAVI AVDHUT PARISH

CO1: How well do you feel this course helped you achieve: "understand the fundamental concepts of algorithm design, performance analysis, and asymptotic notations"?

1 2 3 4 5

Not at all — Completely

CO2: How well do you feel this course helped you achieve: "develop the ability to analyse the time and space complexity of various algorithms and recurrence relations."?

1 2 3 4 5

Not at all — Completely

CO3: How well do you feel this course helped you achieve: "apply divide-and-conquer techniques for solving computational problems efficiently."?

1 2 3 4 5

Not at all — Completely

CO4: How well do you feel this course helped you achieve: "design and implement algorithms using greedy, dynamic programming, backtracking, and branch & bound strategies."?

1 2 3 4 5

Not at all — Completely

CO5: How well do you feel this course helped you achieve: "understand and apply string matching algorithms for efficient pattern searching in real-world applications"?

1 2 3 4 5

Not at all — Completely

CO6: How well do you feel this course helped you achieve: "gain knowledge of computational complexity classes such as P, NP, NP-Hard, and NP-Complete and their significance in problem solving"?

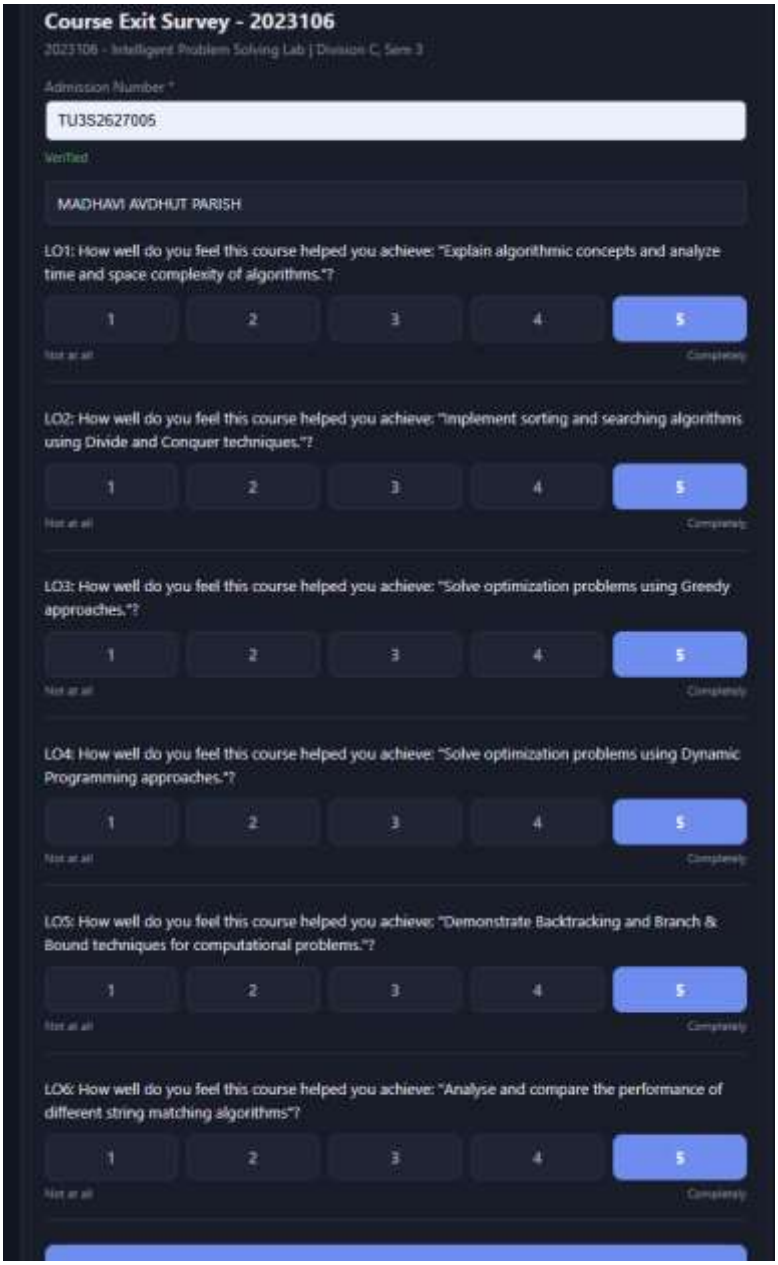
1 2 3 4 5

Not at all — Completely

Submit Survey

Thank you!

Your response has been recorded.


Course Exit Survey - 2023106
2023106 - Intelligent Problem Solving Lab | Division C, Sem 3
Admission Number *
TU3S2627005
Verified
MADHAVI AVDHUT PARISH
LO1: How well do you feel this course helped you achieve: "Explain algorithmic concepts and analyze time and space complexity of algorithms."?
1
2
3
4
5
Not at all
Completely
LO2: How well do you feel this course helped you achieve: "Implement sorting and searching algorithms using Divide and Conquer techniques."?
1
2
3
4
5
Not at all
Completely
LO3: How well do you feel this course helped you achieve: "Solve optimization problems using Greedy approaches."?
1
2
3
4
5
Not at all
Completely
LO4: How well do you feel this course helped you achieve: "Solve optimization problems using Dynamic Programming approaches."?
1
2
3
4
5
Not at all
Completely
LO5: How well do you feel this course helped you achieve: "Demonstrate Backtracking and Branch & Bound techniques for computational problems."?
1
2
3
4
5
Not at all
Completely
LO6: How well do you feel this course helped you achieve: "Analyse and compare the performance of different string matching algorithms"?
1
2
3
4
5
Not at all
Completely


Thank you!
Your response has been recorded.

AIPS Quiz

Total points 0/0

SE-B DIV

Roll No. *

60

Name *

Avdhut Parish Madhavi

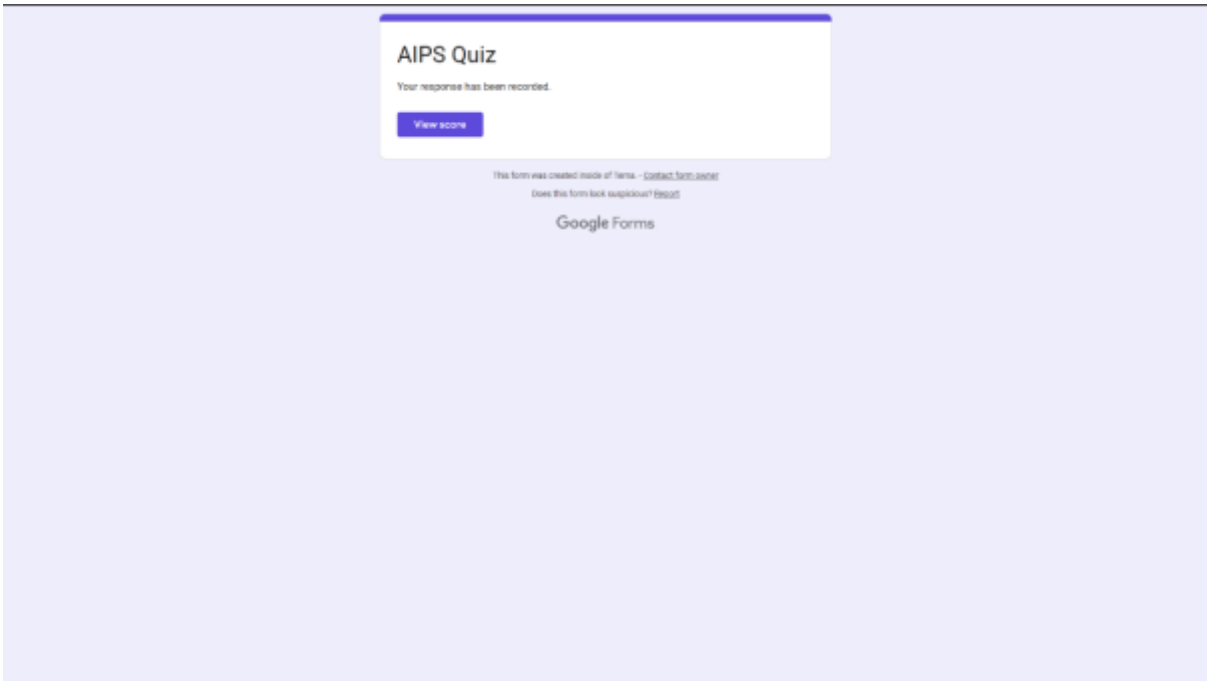
Div *

c

1. What does time complexity measure in an algorithm? *

- (●) Number of operations executed
- ( ) Amount of memory used
- ( ) Number of lines of code
- ( ) Input size

2. What is the space complexity of an algorithm that creates a new array of size n? *


AIPS Quiz
Your response has been recorded.
View score
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Google Forms